Chapter 2. Limits and Continuity

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Contents

2.1 Limits of a function

Definition 1 (Limits of a function)

We write $\lim_{x \to a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a, equals L" if we can make

the values of $f(x)$ arbitrarily close to L

by taking x to be sufficiently close to a (on either side of a) but not equal to a.

An alternative notation for $\lim_{x \to a} f(x) = L$ is

$$f(x) \to L \quad \text{as} \quad x \to a.$$

which is usually read "$f(x)$ approaches L as x approaches a."

Example 2

Estimate the value of

a. $\lim\limits_{x \to 2} (x^2 - x + 2)$ **b.** $\lim\limits_{x \to 2} \dfrac{x^2 - 3x + 2}{x - 2}$ **c.** $\lim\limits_{x \to 0} \dfrac{\sqrt{x^2 + 9} - 3}{x^2}$

Example 3 (Limit that does not exist)

Investigate the limit $\lim\limits_{x \to 0} \sin \dfrac{\pi}{x}$.

One-sided limits

Example 4 (Oliver Heaviside (1850–1925))

The Heaviside function H is defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases} \tag{1}$$

We write

$$\lim_{t \to 0^-} H(t) = 0 \quad \text{and} \quad \lim_{t \to 0^+} H(t) = 1. \tag{2}$$

The symbol

- "$t \to 0^-$" means that we consider only $t < 0$.
- "$t \to 0^+$" means that we consider only $t > 0$.

Definition 5 (One-sided limits)

We write $\lim_{x \to a^-} f(x) = L$ and say "the left-hand limit of $f(x)$, as x approaches a from the left, is equal to L" if we can make

the values of $f(x)$ arbitrarily close to L

by taking x to be sufficiently close to a but x less than a.

Similarly, we can define the right-hand limit of $f(x)$

$$\lim_{x \to a^+} f(x) = L. \tag{3}$$

Theorem 6 (One-sided limit theorem)

$$\lim_{x \to a} f(x) = L \quad \textit{if and only if} \quad \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L \tag{4}$$

Infinity limits

A function f that increases or decreases without bound as x approaches a is said to **tend to infinity** at a.

$$\lim_{x \to a} f(x) = \infty \tag{5}$$

Example 7 (Infinity limits)

Find $\lim_{x \to 0} \frac{1}{x^2}$ if it exists.

Limits at Infinity; Horizontal Asymptotes

Example 8

Find $\lim_{x\to\infty} f(x)$, where

$$f(x) = \frac{x^2 - 1}{x^2 + 1}.$$

As x grows larger and larger, the values of $f(x)$ get closer and closer to 1. We write

$$\lim_{x\to\infty} \frac{x^2 - 1}{x^2 + 1} = 1.$$

2.2 Algebraic computation of limits

1. $\lim\limits_{x \to a} [f(x) \pm g(x)] = \lim\limits_{x \to a} f(x) \pm \lim\limits_{x \to a} g(x)$
2. $\lim\limits_{x \to a} [cf(x)] = c \lim\limits_{x \to a} f(x)$
3. $\lim\limits_{x \to a} [f(x)g(x)] = \lim\limits_{x \to a} f(x) \cdot \lim\limits_{x \to a} g(x)$
4. $\lim\limits_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x \to a} f(x)}{\lim\limits_{x \to a} f(x)}$ if $\lim\limits_{x \to a} g(x) \neq 0$.
5. $\lim\limits_{x \to a} [f(x)]^p = \left[\lim\limits_{x \to a} f(x)\right]^p$ if $p > 0$ and $f(x) > 0$ for x near a.

Example 9

Evaluate the following limits

a. $\lim\limits_{x \to 1} \left(2x^2 - 3x + 4\right)$ **b.** $\lim\limits_{x \to -1} \dfrac{x^3 + 2x^2 - 1}{5 - 3x}$ **c.** $\lim\limits_{x \to 0} \dfrac{x}{\cos x}$

Example 10

Evaluate the following limits

a. $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ **b.** $\lim\limits_{h \to 0} \dfrac{(3 + h)^2 - 9}{h}$ **c.** $\lim\limits_{t \to 0} \dfrac{\sqrt{t^2 + 9} - 3}{t^2}$

Example 11

Find $\lim_{x \to 1} f(x)$ where

$$f(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$$

Example 12

Find $\lim_{x \to 4} f(x)$ if it exists, where

$$f(x) = \begin{cases} \sqrt{x - 4} & \text{if } x > 4 \\ 8 - 2x & \text{if } x < 4 \end{cases}$$

Example 13

Find $\lim_{x \to 3} f(x)$ if it exists

$$f(x) = \begin{cases} 2(x+1) & \text{if } x < 3 \\ 4 & \text{if } x = 3 \\ x^2 - 1 & \text{if } x > 3 \end{cases}$$

Theorem 14 (Special limits involving sine and cosine)

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \quad \textit{and} \quad \lim_{x\to 0} \frac{\cos x - 1}{x} = 0 \tag{6}$$

Example 15

Find each of the following limits

(a) $\displaystyle\lim_{x\to 0} \frac{\sin 3x}{5x}$ $\qquad$ (b) $\displaystyle\lim_{x\to 0} \frac{\sin^{-1} x}{x}$ $\qquad$ (c) $\displaystyle\lim_{h\to 0} \frac{\cos h - 1}{h^2}$

Example 16

Evaluate each of the following limits

$$(a)\ \lim_{t\to 0}\frac{\tan 5t}{\tan 2t} \qquad (b)\ \lim_{x\to 0}\frac{1-\cos x}{\sin x} \qquad (c)\ \lim_{x\to 0}\frac{x^2\cos 2x}{1-\cos x}$$

Theorem 17 (The Squeeze Theorem)

If $f(x) \leq g(x) \leq h(x)$ *when* x *near* a *and*

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then $\lim_{x \to a} g(x) = L$.

Example 18

Show that $\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0$.

2.3 Continuity

Definition 19 (Continuity)

A function f is **continuous at a number** a if

$$\lim_{x \to a} f(x) = f(a) \tag{7}$$

We say that f is **discontinuous at** a (or f has a **discontinuity** at a) if f is not continuous at a.

Notice that Definition 19 implicitly requires three things if f is continuous at a:

- $f(a)$ is defined
- $\lim_{x \to a} f(x)$ exists
- $\lim_{x \to a} f(x) = f(a)$

Example 20

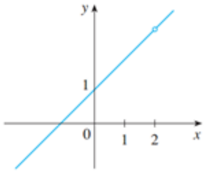
Where are each of the following functions discontinuous?

$$(a)\ f(x) = \frac{x^2 - x - 2}{x - 2}$$

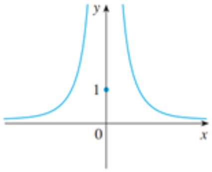
$$(b)\ f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

$$(c)\ f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

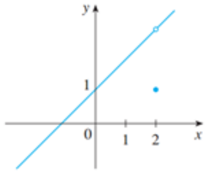
$$(d)^*\ f(x) = [x]$$



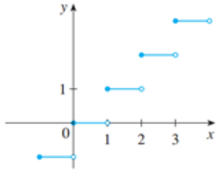
(a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$



(b) $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$



(c) $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$



(d) $f(x) = [\![x]\!]$

Definition 21 (One-sided Continuity)

A function f is **continuous from the right at a number** a if

$$\lim_{x \to a^+} f(x) = f(a) \tag{8}$$

and f is **continuous from the left at** a if

$$\lim_{x \to a^-} f(x) = f(a) \tag{9}$$

Example 22

Show that the function $f(x) = [x]$ is continuous from the right but discontinuous from the left at each integer n.

Definition 23 (Continuity on an interval)

A function f is **continuous on an interval** (a, b) if it is continuous at every number in the interval. Moreover

- If f is continuous from the right at a, then we say it is **continuous on the half-open interval** $[a, b)$.
- If f is continuous from the left at b, then we say it is **continuous on the half-open interval** $(a, b]$.
- If f is continuous from the right at a and continuous from left at b, then we say it is **continuous on the closed interval** $[a, b]$.

Example 24

Find the intervals on which each following function is continuous

$$(a)\ f(x) = \frac{x^2 - 1}{x^2 - 4} \qquad (b)\ f(x) = \left|x^2 - 4\right| \qquad (c)\ f(x) = \sin\frac{1}{x}$$

Example 25

Find the intervals on which each following function is continuous

a. $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$ **b.** $g(x) = \begin{cases} 3 - x & \text{if } \; -5 \leq x < 2 \\ x - 2 & \text{if } \quad 2 \leq x < 5 \end{cases}$

c. $h(x) = \begin{cases} x \sin \dfrac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 26

Find constants a and b so that the given function will be continuous for all x.

a. $f(x) = \begin{cases} x^2 + bx + 1 & \text{if } x < 5 \\ 8 & \text{if } x = 5 \\ ax + 3 & \text{if } x > 5 \end{cases}$ **b.** $f(x) = \begin{cases} b & \text{if } x \leq 1 \\ \dfrac{\sqrt{x} - a}{x - 1} & \text{if } x > 1 \end{cases}$

Theorem 27 (The Intermediate Value Theorem)

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

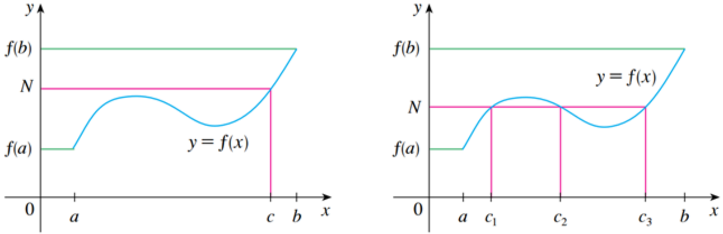


Figure: The Intermediate Value Theorem

Theorem 28 (Root location theorem)

If f is continuous on the closed interval $[a, b]$ and if $f(a)$ and $f(b)$ have opposite algebraic signs, then $f(c) = 0$ for at least one number c in the open interval (a, b).

Example 29

Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

between 1 and 2.

Example 30

Show that $\cos x = x^3 - x$ has at least one solution on the interval $\left(0, \frac{\pi}{2}\right)$.

Example 31

Show that the equation $\sqrt[3]{x} = x^2 + 2x - 1$ has at least one solution on the interval $(0, 1)$.

Exercise 1 (Exercise 54, p. 106)

The population (in thousands) of a colony of bacteria t minutes after the introduction of a toxin is given by the function

$$P(t) = \begin{cases} t^2 + 1 & \text{if } 0 \leq t < 5 \\ -8t + 66 & \text{if } t \geq 5 \end{cases}$$

(a) *When does the colony die out?*

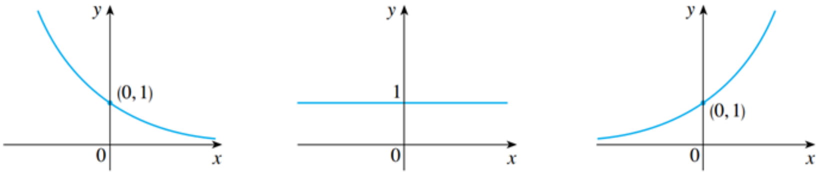
(b) *Show that at some time between $t = 2$ and $t = 7$ the population is 9,000.*

In general, an **exponential function** is a function of the form

$$f(x) = a^x$$

where a is a positive constant.

- if $0 < a < 1$ the exponential function decreases;
- if $a = 1$, it is a constant;
- if $a > 1$, it increases.



(a) $y = a^x$, $0 < a < 1$ (b) $y = 1^x$ (c) $y = a^x$, $a > 1$

Laws of Exponents

If a and b are positive numbers and x and y are any real numbers, then

1. $a^{x+y} = a^x a^y$ **2.** $a^{x-y} = \dfrac{a^x}{a^y}$ **3.** $(a^x)^y = a^{xy}$ **4.** $(ab)^x = a^x b^x$

Problem 1 (Population growth)

First we consider a population of bacteria in a homogeneous nutrient medium. Suppose that by sampling the population at certain intervals it is determined that the population doubles every hour.

If the number of bacteria at time t is $P(t)$, where t is measured in hours, and the initial population is $P(0) = 1000$, then we have

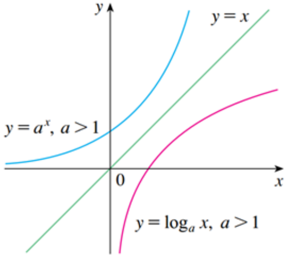
$$P(1) = 2P(0) = 2x1000$$
$$P(2) = 2P(1) = 2^2x1000$$
$$P(3) = 2P(2) = 2^3x1000$$

In general, $P(t) = 2^t x1000 = (1000)2^t$.

The logarithmic function with base a denoted $y = \log_a x$ $(0 < a \neq 1)$ is the inverse function of the exponential function $y = a^x$.

We have that

$$\log_a x = y \iff a^y = x.$$



By cancellation equations, we have

$$\log_a (a^x) = x \quad \text{for every } x \in \mathbb{R}$$
$$a^{\log_a x} = x \quad \text{for every } x > 0$$

Some special bases:

- The **common logarithm**: $\log_{10} x = \log x$
- The **natural logarithm**: $\log_e x = \ln x$ (read "ell n x" or "lawn x")

Laws of Logarithms

If x and y are positive numbers, then

1. $\log_a(xy) = \log_a x + \log_a y$
2. $\log_a\left(\dfrac{x}{y}\right) = \log_a x - \log_a y$
3. $\log_a(x^r) = r\log_a x \quad$ (where r is any real number)

Change of Base Formula

For any positive number a ($a \neq 1$), we have

$$\log_a x = \frac{\ln x}{\ln a}$$

Example 32 (Exponential growth)

A biological colony grows in such a way that at a time t (in minutes), the population is

$$P(t) = P_0 e^{kt}$$

where P_0 is the initial population and k is a positive constant. Suppose the colony begins with $5,000$ individuals and contains a population of 7,000 after 20 min. Find k and determine the population (rounded to the nearest hundred individuals) after 30 min.

The Number e; Natural Growth

Assume that we invest \$1 with the interest rate 100% per year. Compute the profit P when the interest is compounded n times per year, continuously?

- Annual growth: $P = 2$ or $P = 1 + 100\%$
- Semi-annual growth: $P = 2.25$ or

$$P = \left(1 + \frac{100\%}{2}\right)^2$$

- If the interest is compounded n times per year, then

$$P = \left(1 + \frac{100\%}{n}\right)^n$$

  Since the interest is compounded continuously, we have

$$P = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = e \approx 2.718281828...$$

If P dollars are compounded n times per year at an annual rate r, then the future value after t years is given by

$$A(t) = P\left(1 + \frac{r}{n}\right)^{nt}. \tag{10}$$

And if the compounding is continuous, then the future value is

$$A(t) = Pe^{rt} \tag{11}$$

Example 33

If \$12,000 is invested for 5 years at the rate 4%, find the future value at the end of 5 years if interest is compounded:

(a) monthly; continuously.

(b) If the interest is compounded continuously, how long will it take for the money to double?